



Veeam Backup & Replication Malware Detection Demo

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Objective: To demonstrate, in the Veeam Backup & Replication console, some of the malware/ransomware detection capabilities (e.g., Inline Malware Detection, Anomaly/IOC Detection, Yara rule and Threat Hunter) during and after backup runs of Windows and Linux VMs.

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Prerequisites:

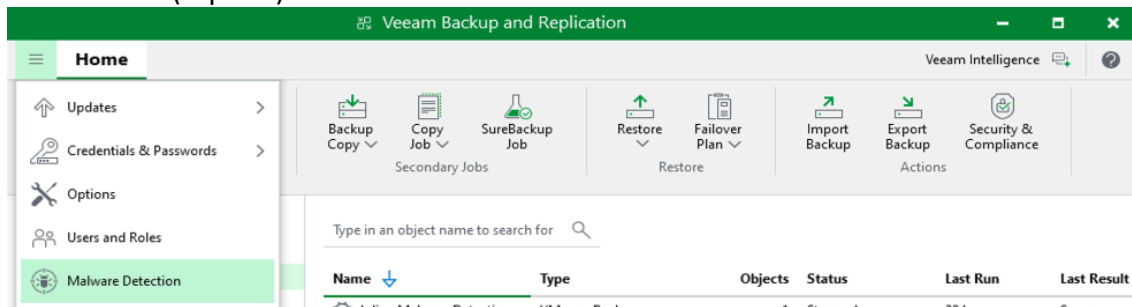
- **Lab environment:** Run this script only in isolated environment (no access to the corporate/production network), with snapshots/checkpoints of the VMs.
- **Access:** Administrative credentials in VBR and guest credentials (Windows/Linux) registered in Veeam (Guest Processing).
- **Veeam Components:** Veeam Backup & Replication installed and functional; repository(s) configured; proxy(s) and hypervisor access (vSphere/Nutanix) configured.
- **Malware Features:** Ensure that the features that will be demonstrated (e.g., Inline Malware Scan / Anomaly Detection / IOC / YARA) are enabled in your environment.

Summary

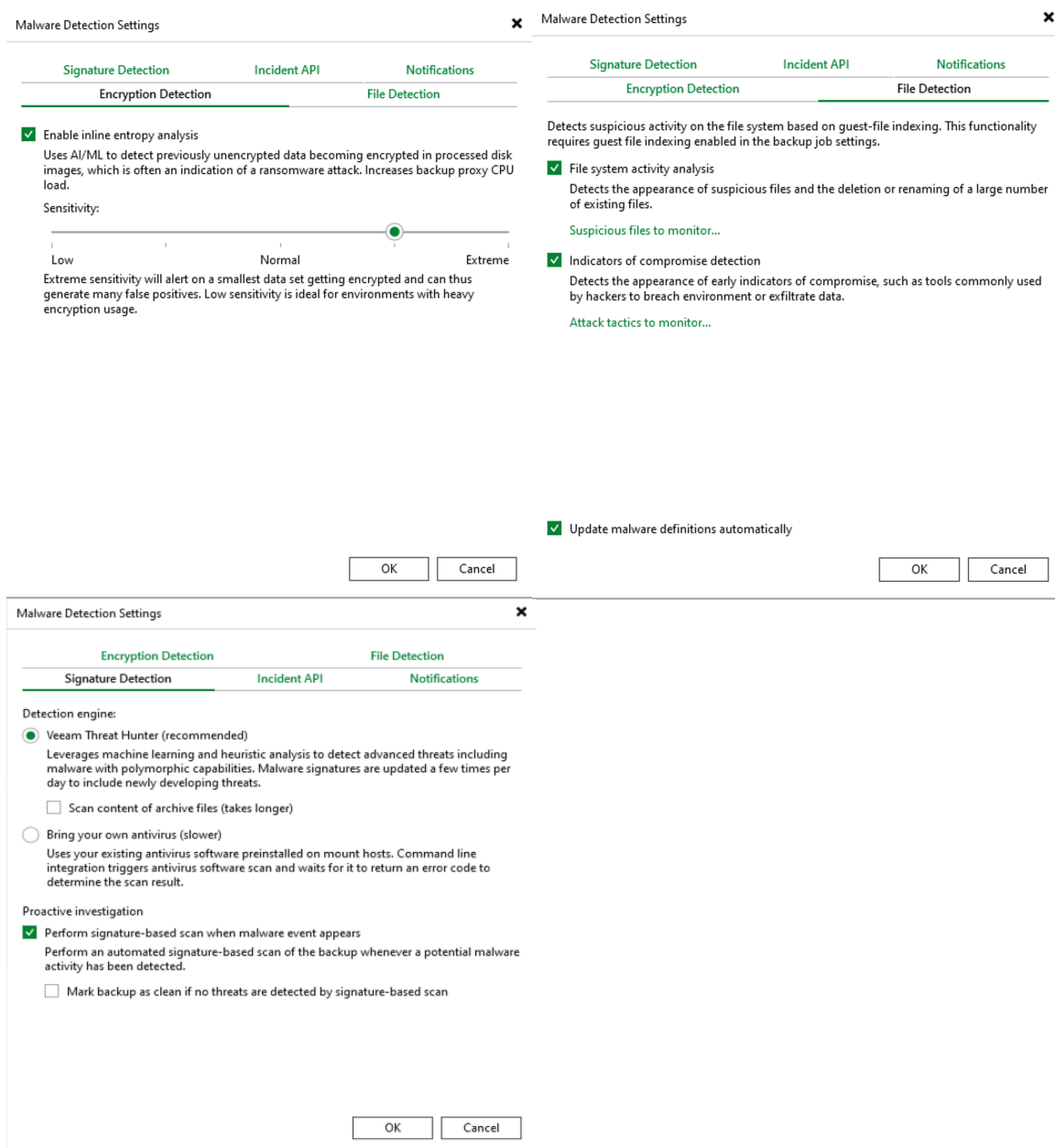
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1. Enabling and Validating Malware Detection Mechanisms

- a) **Open the Veeam Backup & Replication (VBR) console** and validate that the detection mechanisms are enabled.
- b) Go to **Menu (top left) > Malware Detection**



- c) Find the settings related to **Encryption detection**, **File Detection**, and **Signature Detection** and adjust as per the example below:



2. Configuring Backup Jobs and Validating Settings

2.1 Protecting the VMs

- Go to **Home > Jobs → Backup Jobs → Virtual Machine** (Select Hypervisor)
- Set a name [Malware Detection] for the Backup Job and click Next.

The 'New Backup Job' dialog is shown with the 'Name' tab selected. The 'Name' field contains 'Malware Detection'. The 'Description' field is empty.

Tab	Field	Value
Name	Name	Malware Detection
Name	Description	

- Add the two VMs (SRV1 and SRV2) and click Next.

The 'New Backup Job' dialog is shown with the 'Virtual Machines' tab selected. It displays a table of virtual machines to be backed up.

Name	Type	Size
SRV1	Virtual machine	42.3 GB
SRV2	Virtual machine	14.4 GB

Buttons: + Add..., - Remove

- On the Storage tab, click Next.
- In the **Guest Processing** tab, select **Enable guest file system indexing and malware detection** and then click on "Indexing Options for individual machines"

The 'New Backup Job' dialog is shown with the 'Guest Processing' tab selected. The 'Enable guest file system indexing and malware detection' checkbox is checked. The 'Indexing options for individual machines' link is highlighted.

☒ Enable guest file system indexing and malware detection
Indexing enables global file search functionality, automatic detection of suspicious file system activity and known malware files.
[Indexing options for individual machines](#)

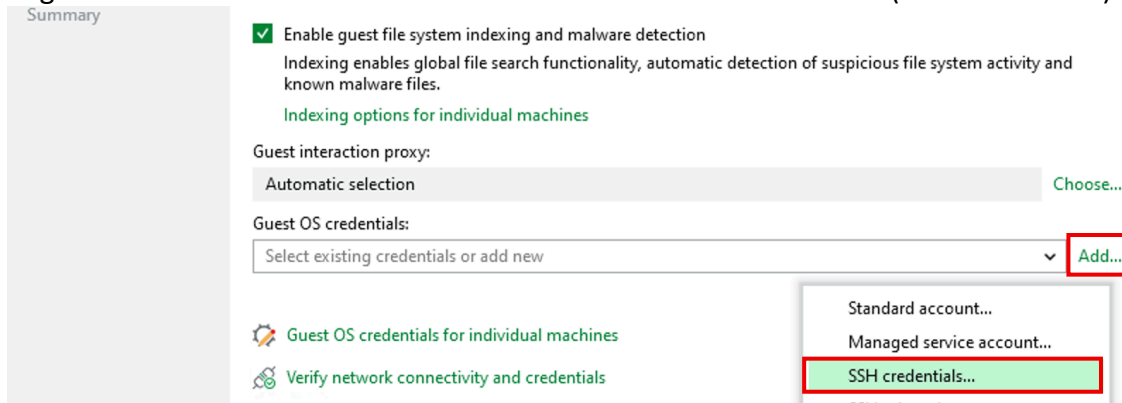
- Edit the **Index Options for individual machines** as follows:

The 'Guest File System Indexing Options' dialog is shown. It displays a table of indexing settings for individual items.

Object	Windows	Linux
SRV1	Disabled	Disabled
SRV2	Disabled	Everything

Buttons: Add..., Edit..., Remove, OK, Cancel

- g) Register the Guest OS credential via the **Add** button for SRV2 server (SSH Credentials)



Summary

☒ Enable guest file system indexing and malware detection
Indexing enables global file search functionality, automatic detection of suspicious file system activity and known malware files.
Indexing options for individual machines

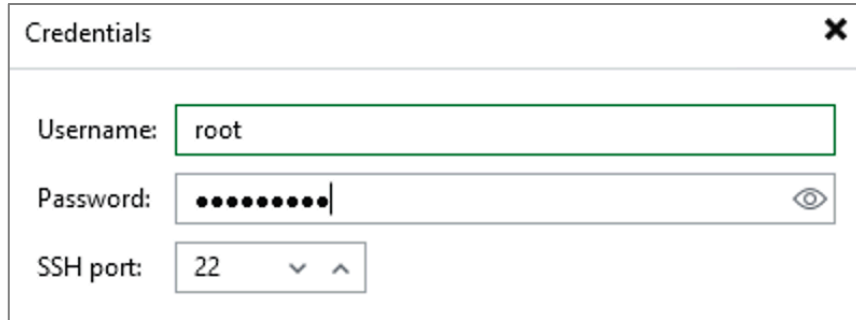
Guest interaction proxy:
Automatic selection Choose...

Guest OS credentials:
Select existing credentials or add new Add...

Guest OS credentials for individual machines
Verify network connectivity and credentials

Standard account...
Managed service account...
SSH credentials...

- h) Enter the following username and password [**root / Veeam123!**] and click OK.



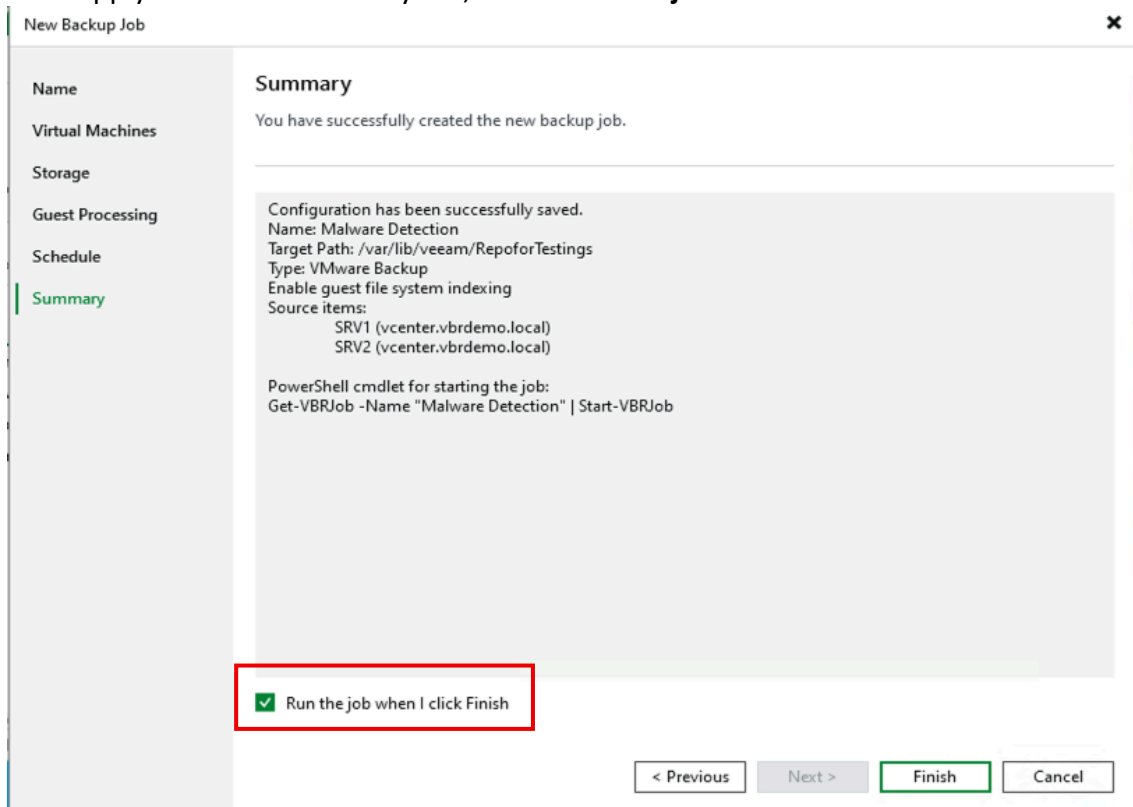
Credentials

Username: root

Password: ●●●●●●●●

SSH port: 22

- i) We will not index the backup of the SRV1 server so we will not need to set a credential for it. *But if you need to do this in a production environment, just use the **Guest OS Credential for individual machines** option.*
- j) Click (**OK / Next**), on the Schedule tab, proceed **without setting a time**.
- k) Click Apply → On the Summary tab, check **Run the job When i click Finish**



New Backup Job

Summary

You have successfully created the new backup job.

Configuration has been successfully saved.
Name: Malware Detection
Target Path: /var/lib/veeam/RepoforTestings
Type: VMware Backup
Enable guest file system indexing
Source items:
SRV1 (vcenter.vbrdemo.local)
SRV2 (vcenter.vbrdemo.local)

PowerShell cmdlet for starting the job:
Get-VBRJob -Name "Malware Detection" | Start-VBRJob

☒ Run the job when I click Finish

< Previous Next > Finish Cancel

2.2 Creating the Encryption Baseline/Anomalies

- Track the **progress of the job** in Last 24 Hours (**or in Sessions**).
- When finished or during backup, open the **Statistics / Details** of the job and look for indications of:
- Guest indexing** (Linux VM – SRV2)

Name	Status	Action	Duration
SRV2	Success	Processing SRV2	0:03:26
		VM processing started at 3/9/2026 1:59:37 PM	
		VM size: 36 GB (14.5 GB used)	
		Resetting CBT per job settings for active fulls	0:00:06
		Getting VM info from vSphere	0:00:07
		Using guest interaction proxy vbr1.vbrdemo.local (Same subnet)	
		Creating VM snapshot	0:00:02
		Indexing guest file system	
		Saving [EX1-NVME1] SRV2/SRV2.vmx	
		Saving [EX1-NVME1] SRV2/SRV2.nvram	

Hide details OK

- Inline Malware Scan (Windows VM – SRV1)

Name	Status	Action	Duration
SRV1	Success	Saving [EX1-NVME2] SRV1/SRV1.nvram	
		Using backup proxy prx1.vbrdemo.local for disk Hard disk 2 [hotadd]	0:00:12
		Using backup proxy prx1.vbrdemo.local for disk Hard disk 1 [hotadd]	0:00:23
		Hard disk 2 (25 GB) 21 GB read at 391 MB/s [CBT]	0:00:55
		Hard disk 1 (80 GB) 15.6 GB read at 330 MB/s [CBT]	0:00:50
		Removing VM snapshot	0:00:02
		Inline malware detection increases backup proxy CPU usage and may affect processin...	
		Publishing malware detection metadata to catalog	
		Finalizing	
		Swap file blocks skipped: 39 MB	

Hide details OK

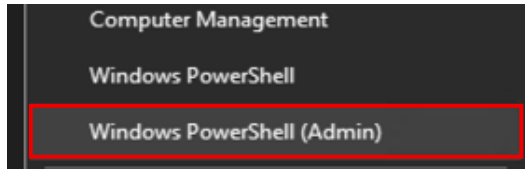
This backup procedure can take between 5-10 minutes.

ATTENTION: BEFORE PROCEEDING TO THE NEXT STEP, IT IS MANDATORY TO WAIT FOR THE FULL BACKUP OF BOTH VMS TO COMPLETE!

3. Simulating encryption of a Windows server

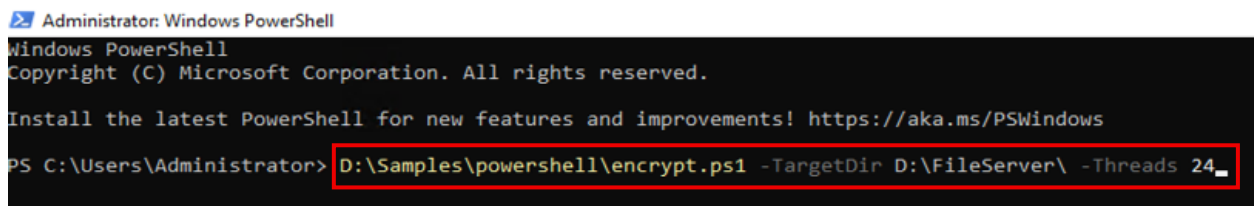
Using Powershell script we will encrypt the files using AES 256 and create fake onion links in each infected directory. The process aims to simulate a malicious encryption of a file server.

- Access **SRV1** by console or through the RDP (if enabled)
- Enter the following username and password [**Administrator / Veeam123!**]
- Run a PowerShell session as an administrator.



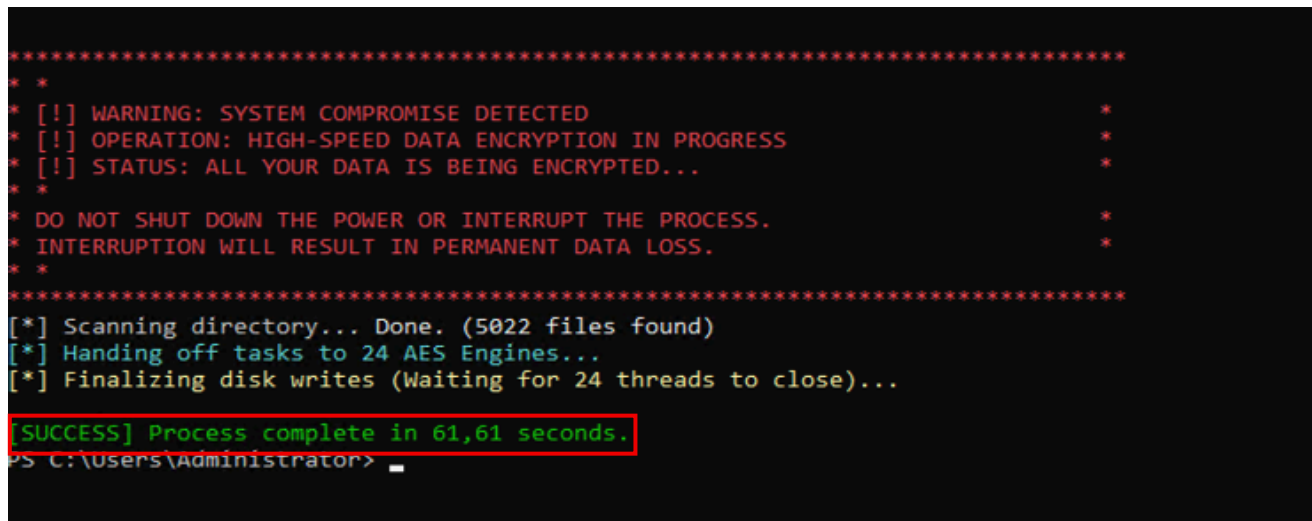
- In PowerShell, type the following command:

```
D:\Samples\powershell\encrypt.ps1 -TargetDir D:\FileServer\ -Threads 24
```



Note: The number of threads varies according to the number of CPU cores; In general, each core can run 2 threads (number of cores $\times$ 2 = number of threads).

- Wait for the script to complete or go to step 4 (e.g., script completed).



4. Linux Server Anomaly Simulation

On the Linux server, run the rename script from the /data/fileserver/rename.sh directory

- Access the Linux server SRV2 (SSH or console). [**root / Veeam123!**]
- Go to the **/data/fileserver directory** that will be changed and run the rename.sh script:

```
cd /data/fileserver
```

```
./rename.sh
```

```
Last login: Mon Mar  9 16:57:47 2026 from 10.10.1.5  
[root@srv2 ~]# cd /data/fileserver/  
[root@srv2 fileserver]# ./rename.sh
```

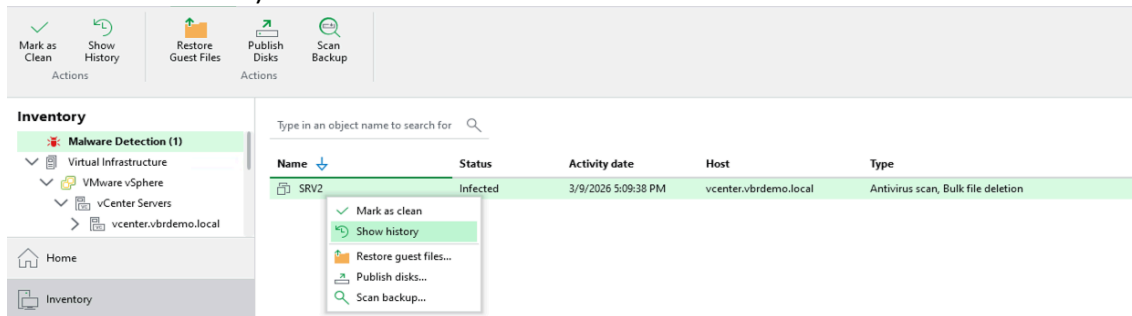
- Wait for the script to complete, access a subdirectory and you will be able to notice that all files now have the extension [xpto], through the command

CD Files && ll

```
[root@srv2 fileserver]# cd files && ll  
total 3208  
drwxr-xr-x. 8 root root    150 Feb 23 18:10 Accounting  
-rw-r--r--. 1 root root  35067 Jun  5  2022 confCons.xml  
-rw-r--r--. 1 root root  35067 Jun  5  2022 confCons.xml.20220605-1557253669.backup  
-rw-r--r--. 1 root root 469513 Jun 30  2023 file-example_PDF_500_kB.pdf  
-rw-r--r--. 1 root root 1028608 Jun 30  2023 file_example_PPT_1MB.ppt  
-rw-r--r--. 1 root root  672256 Jun 30  2023 file_example_XLS_5000.xls  
-rw-r--r--. 1 root root 1027072 Jun 30  2023 file-sample_1MB.doc  
drwxr-xr-x. 7 root root    81 Feb 23 18:13 Finance  
drwxr-xr-x. 9 root root   111 Feb 23 18:12 IT  
-rw-r--r--. 1 root root   1502 Sep 21  2023 license-main-1000.BETA1.lic  
drwxr-xr-x. 7 root root    81 Feb 23 18:11 Manufacturing  
drwxr-xr-x. 7 root root    81 Feb 23 18:11 Marketing  
drwxr-xr-x. 7 root root    81 Feb 23 18:11 'R&D'  
drwxr-xr-x. 2 root root   120 Feb 23 18:11 RHEL  
drwxr-xr-x. 7 root root    81 Feb 23 18:11 Sales  
drwxr-xr-x. 3 root root    18 Feb 23 18:10 veeam_downloads
```


5. Detecting File System Anomalies

- Under **Home > Jobs**, select the job that protects the VMs, and then click **Start**.
- Once complete, navigate to **Inventory** and locate the **Malware Detection** view, select the **SRV2 server**, and choose **Show History** (note that *Bulk file deletion* was detected and this triggered a *Threat Hunter scan*).



- In the **History** section, it is possible to identify which file types were renamed:

Event Details ✕

Object:	SRV2
Event created:	3/9/2026 5:10 PM
Activity date:	3/9/2026 5:09 PM
Event type:	Bulk file deletion
Initiated by:	veeam-usr-vbr
Status:	Suspicious

Details:

Locations of deleted or renamed files can be found on the backup server vbr1.vbrdemo.local at /var/log/VeeamBackup/Malware_Detection_Logs/deleted_files_SRV2_2026-03-09-20-09-38_2026-03-09-17-02-10.log

Potential malware activity detected: too many .docx,.xlsx,.ppt,.rtf,.pdf,.jpg,.png,.gif,.mp4,.avi,.wmv,.mp3,.zip files have been deleted or renamed since last backup, ensure this activity is legitimate

- Note that this triggered a Threat Hunter-type session that identified **EICAR** (our fake malware) inside the file system:

Event Details ✕

Object:	SRV2
Event created:	3/9/2026 5:11 PM
Activity date:	3/9/2026 5:09 PM
Event type:	Antivirus scan
Initiated by:	veeam-usr-vbr
Status:	Infected

Details:

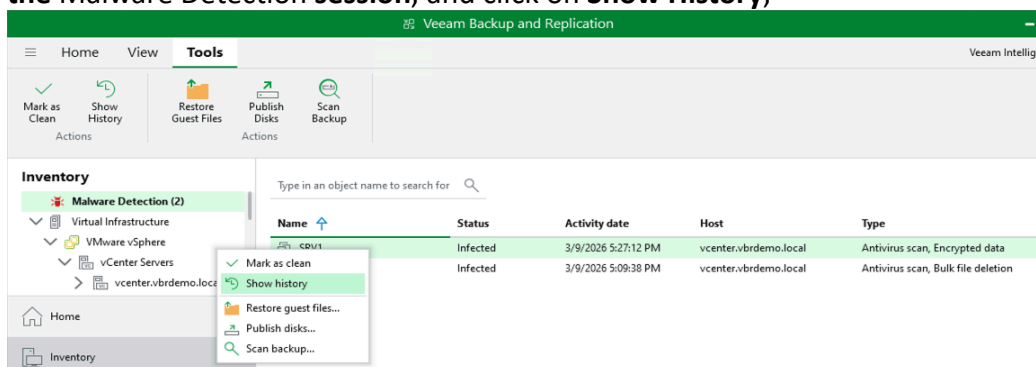
Malware detected by Veeam Threat Hunter, see antivirus scan log on mount server [vbr1.vbrdemo.local] in [FLRSessions/OtherOS/FLR_SRV2_/Antivirus] /data: Infected

6. Detecting encryption on servers

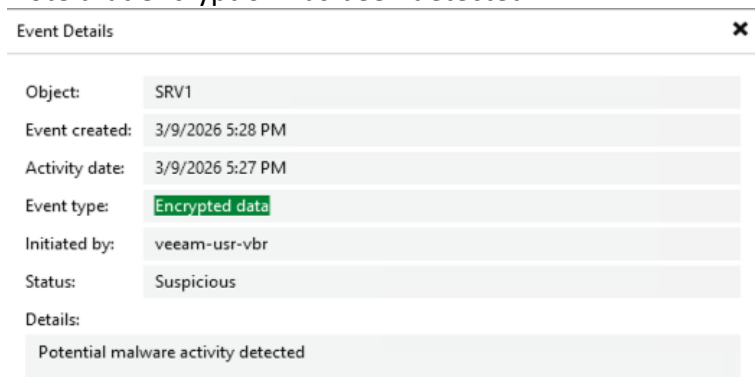
- a) Verify that the server encryption process has finished.

```
*****
* *
* [!] WARNING: SYSTEM COMPROMISE DETECTED
* [!] OPERATION: HIGH-SPEED DATA ENCRYPTION IN PROGRESS
* [!] STATUS: ALL YOUR DATA IS BEING ENCRYPTED...
* *
* DO NOT SHUT DOWN THE POWER OR INTERRUPT THE PROCESS.
* INTERRUPTION WILL RESULT IN PERMANENT DATA LOSS.
* *
*****
[*] Scanning directory... Done. (5022 files found)
[*] Handing off tasks to 24 AES Engines...
[*] Finalizing disk writes (Waiting for 24 threads to close)...
[SUCCESS] Process complete in 52,52 seconds.
```

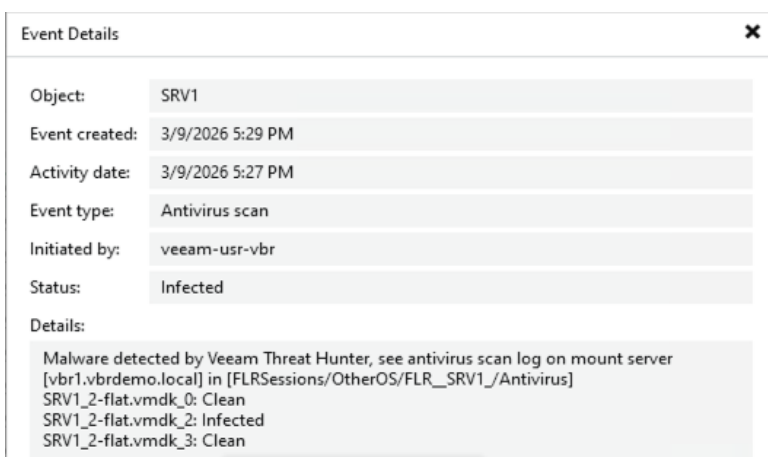
- b) Run the Backup Job again and wait for it to complete.
c) When the backup job is finished, go to the **Inventory** tab and locate the Windows **SRV1** server in the Malware Detection session, and click on **Show History**;



- d) Note that encryption has been detected:



- e) This triggered an antivirus scan (*Threat Hunter*) that detected malware in the backup:



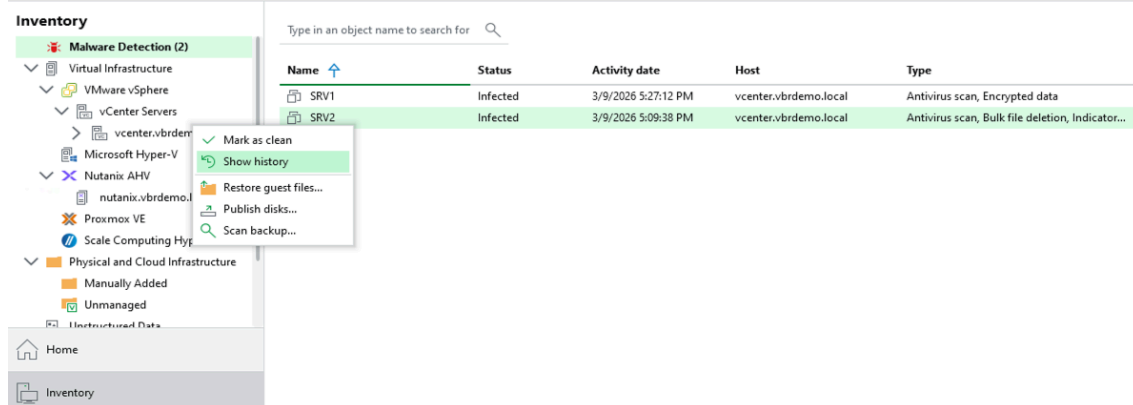
7. Indicators of Compromise – IOC [DEMO]

In Veeam Backup & Replication, Indicators of Compromise (IoCs) are executables whose unexpected presence on a system may indicate security compromise

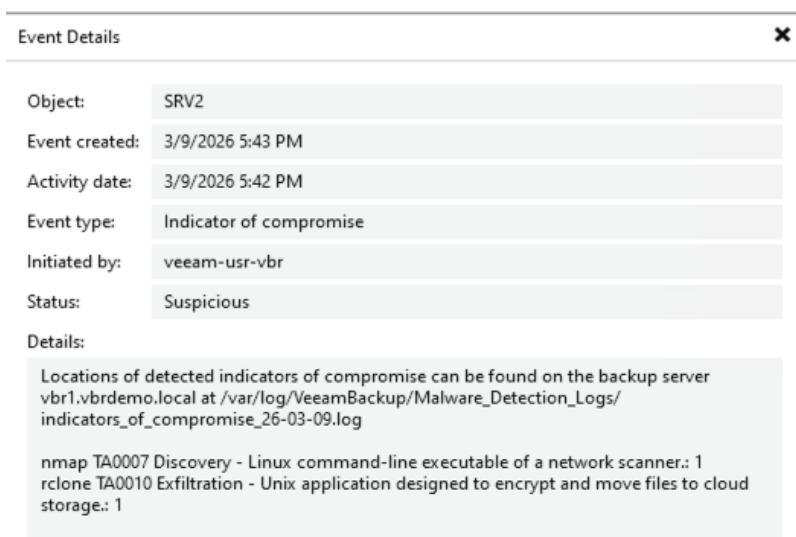
The identification of these IoCs is based on the list of executables defined by MITRE ATT&CK.

https://helpcenter.veeam.com/docs/vbr/userguide/malware_detection_guest_index_ioc.html?ver=13

- Access the Linux server via SSH. [root/Veeam123!]
- Install **nmap** using the YUM package manager:
`yum install -y nmap`
- Install **rclone** as per procedure below.
`curl https://rclone.org/install.sh | sudo bash`
- Perform the incremental backup again.
- Open the server's **Malware Detection** history again (note that it has now been cited *Indicators...*).



- View the details of the session that encountered the **Indicator of Compromise**:



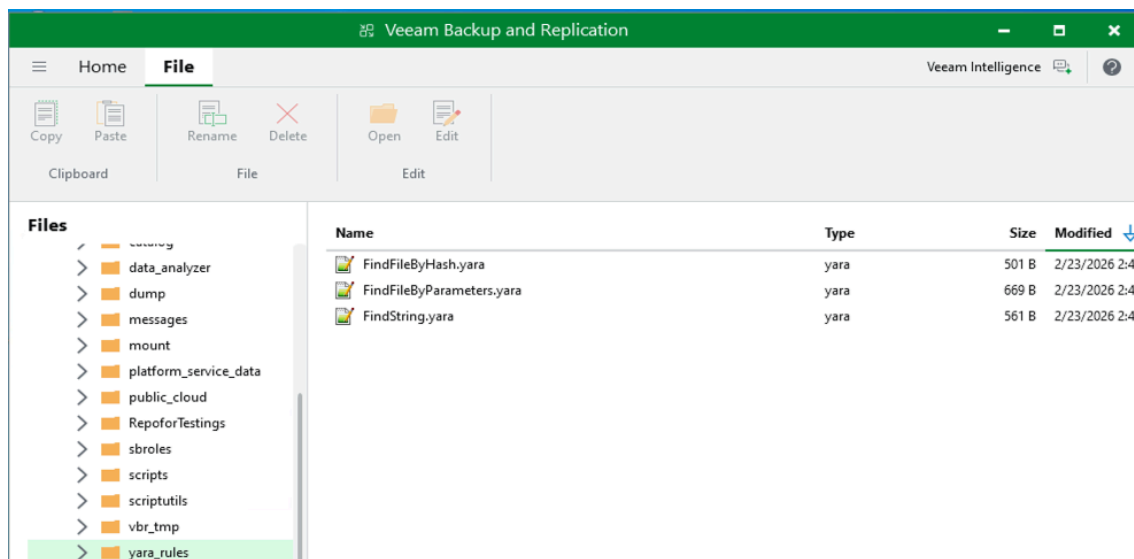
8. Demonstrating integration with YARA Rules

YARA rules are especially useful when we need to look for patterns in binaries or files that do not yet have a known extension or a signature recognized by traditional antivirus engines. In these situations, we can create rules that analyze the content of the file, its size, hash, among several other characteristics. This allows you to locate malware or any other relevant data, even if they don't have a known signature yet, making detection much more flexible and effective.

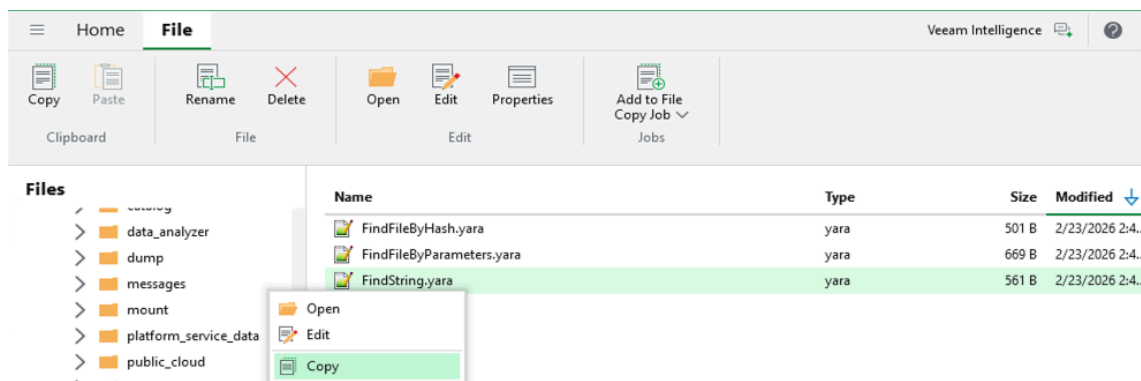
a) Create a rule as per the example below and import it into VBR:

Tip: You can use ChatGPT or another LLM to quickly create YARA rules for any purpose.

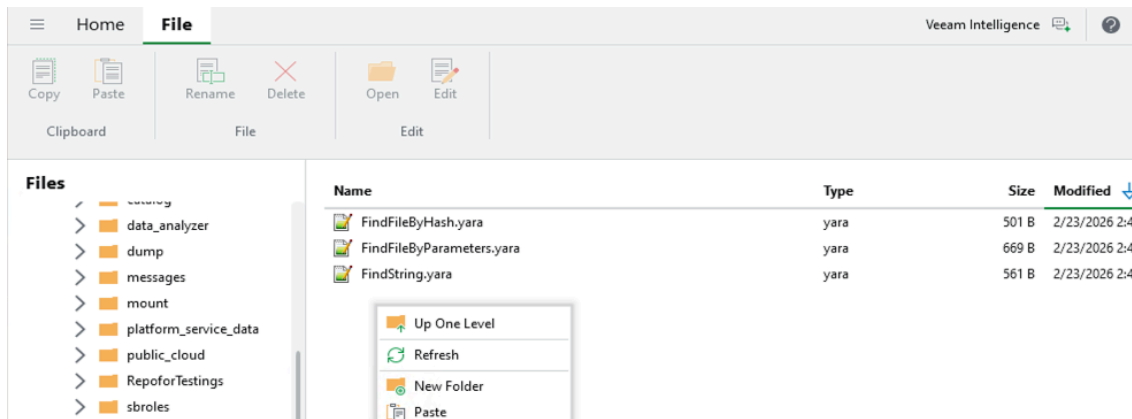
b) Go to the Files menu of the VBR → This Server → Yara Rules:



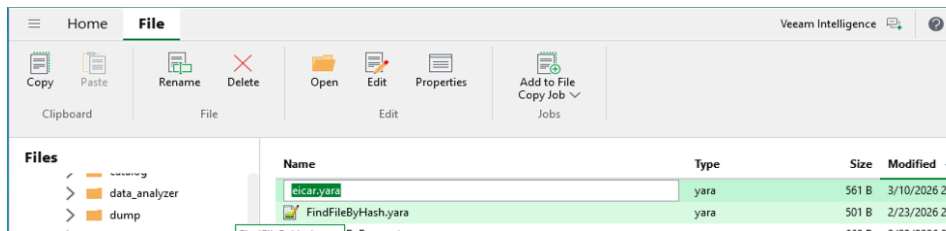
c) Select the **FindString.yara** rule → Right click and Copy



d) Right-click the empty area Paste →



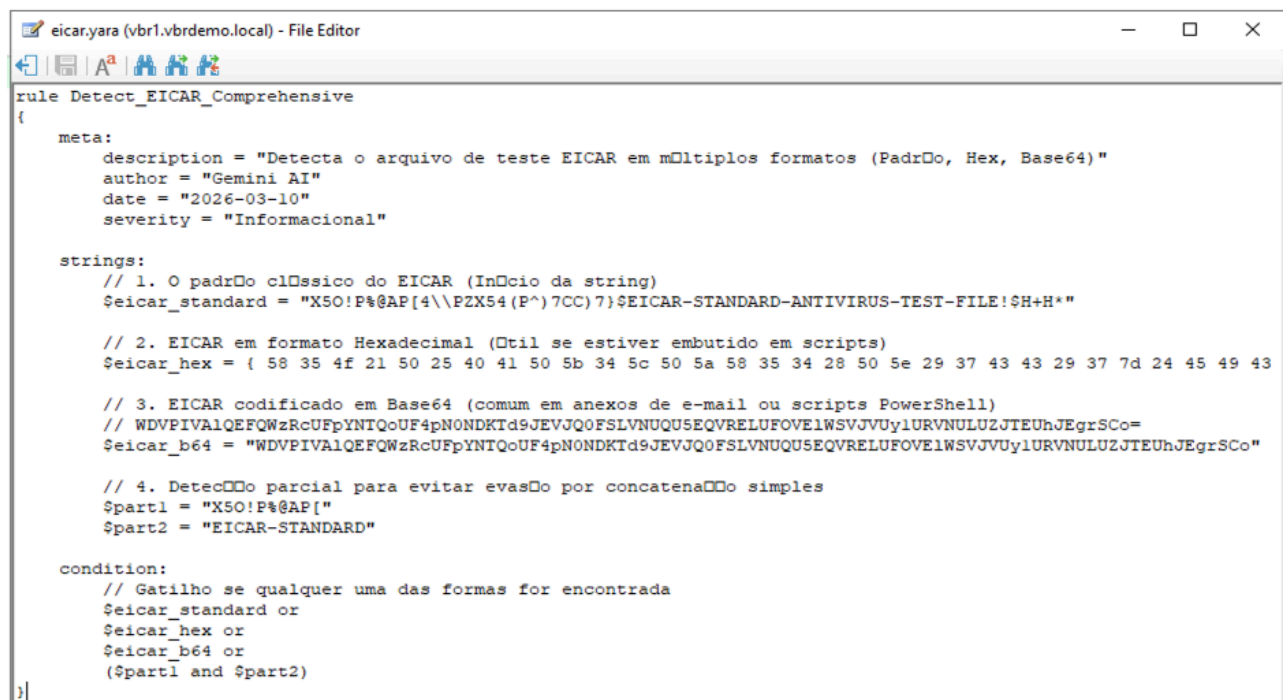
e) Select the rule "Copy of FindString.yara" → Rename → "eicar.yara"



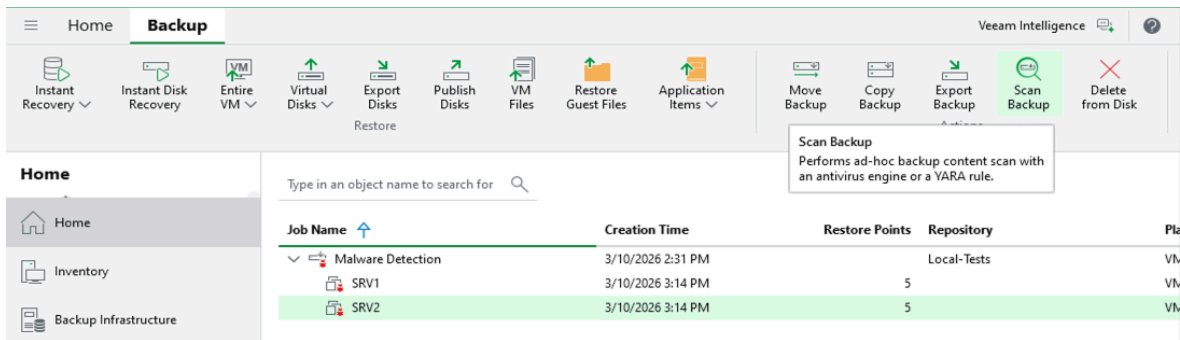
f) Click the Edit button and paste the content of link below into the Yara rule content [eicar.yara].

<https://github.com/magnunscheffer/veeam/blob/main/demo-malware/eicar.yara>

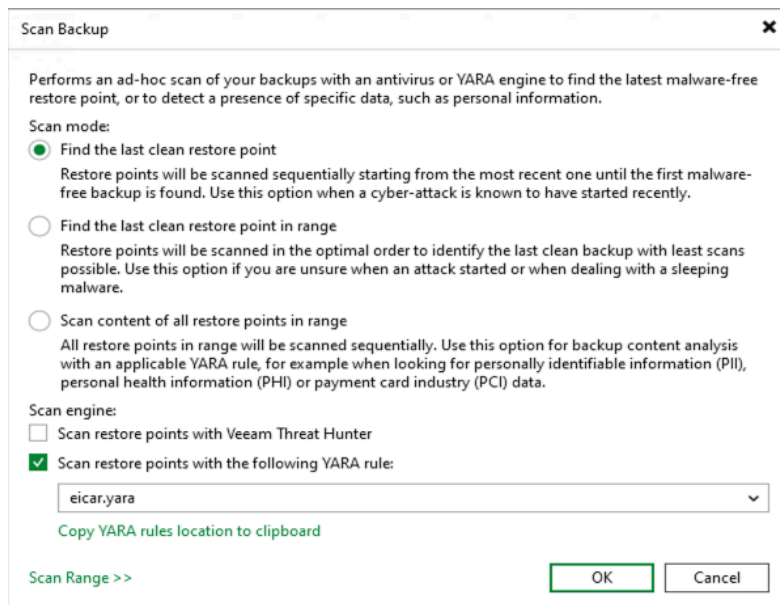
g) Click the **Save** icon and then close the text editor:



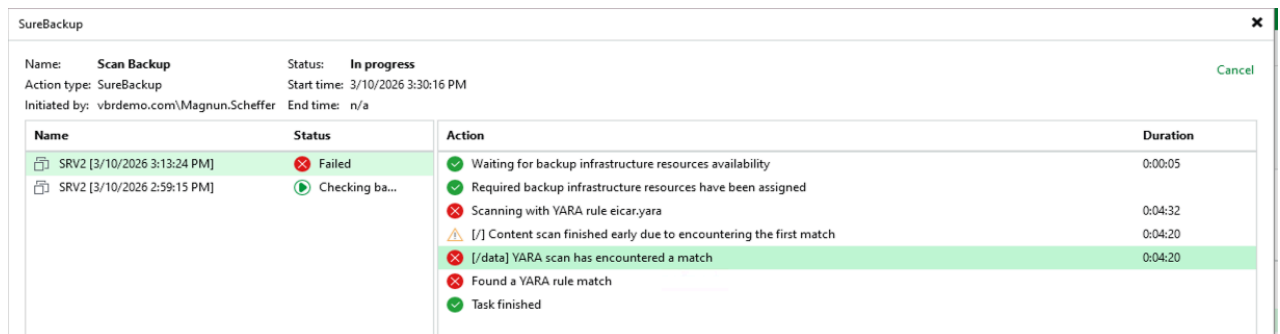
h) After loading the Yara rule, let's simulate the location of the **eicar** files (dummy malware), locate the backups from the SRV2 server (Linux) in the Home → Backups → Disk → SRV2 menu, click on **Scan Backup** in the top bar or via right click.



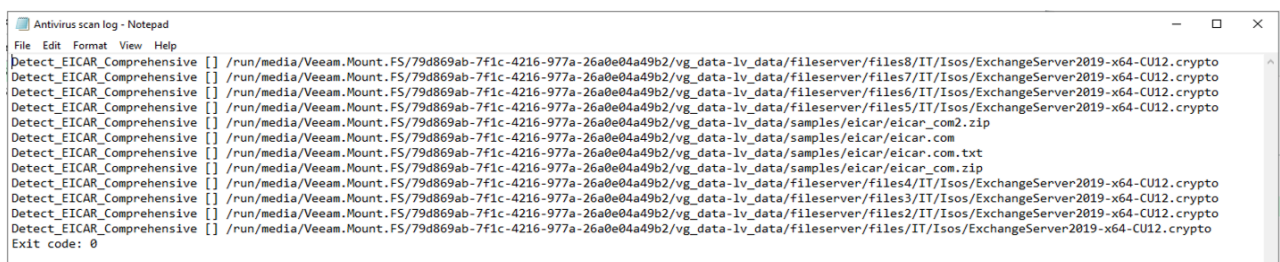
i) Select the **Scan** option.... **YARA rule:** → Choose the rule you created earlier and press **OK**



j) Wait for the failure to happen, if a failure occurs it means that we had a match, that is, malware was identified at that recovery point compatible with the standard configured in the Yara rule.



k) You can open the **Scan Log** to show in detail where the match was located:



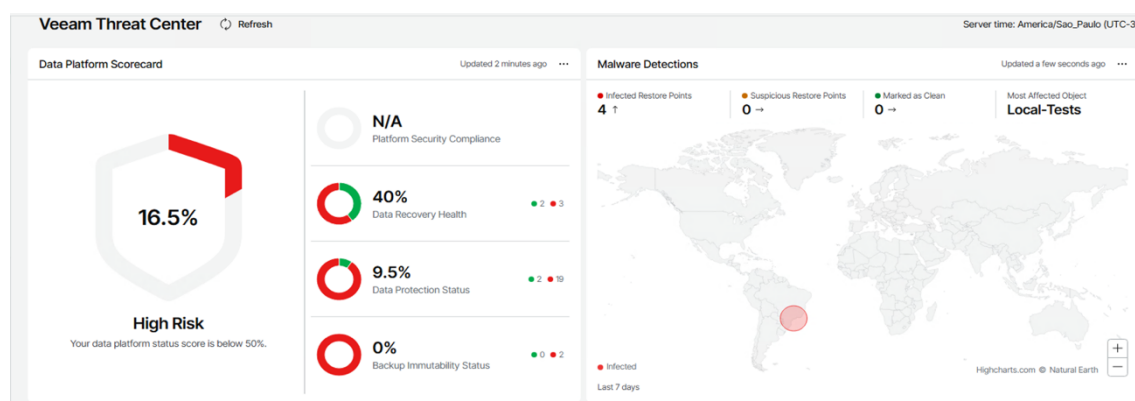
9. Visualizing the situation via Threat Center (Not available in this lab)

If your lab has Veeam One, you can obtain a lot of relevant information regarding these mechanisms demonstrated above, such as:

- Scoreboard of the backup environment affected by malware
- Alarms with programmable actions (e.g. shutdown of the VM via script, send a notification to the SIEM, open a ticket in servicenow).
- Reports detailing malware activities are also available.

See some examples below:

a) Threat Center dashboard and how the score was affected



b) Malware alarms

Selected Object: vbr1.vbrdemo.local (version: 13.0.1.1071) - 2 errors, 2 warnings

Status	Time	Source	Type	Name
Error	18:02:24	This object (vbr1.vbrdemo.local)	Potential malware in backups	
Error	18:02:24	vSphere VM "SRV2"		vSphere VM "SRV2" has been infected Detection source: Internal malware scan Additional details: Malware detected by Veeam Threat Hunter, s vSphere VM "SRV2" has been infected Detection source: Internal malware scan ...
Error	17:31:21	vSphere VM "SRV1"		vSphere VM "SRV1" has been infected Detection source: Internal malware scan Additional details: Malware detected by Veeam Threat Hunter, s Suspicious malware activity was detected on vSphere VM "SRV1" Detection source: Internal malware scan ...

c) Programmable actions such as forwarding this type of alarm to a specific SIEM tool or plugin.

Alarm Settings

Notifications are sent when alarms with the corresponding severity are triggered. Use the list below to define notification options.

Action	Value	Condition	
Send email to a default g...		Any state	Add
Send email to a default group			Remove
Send email notification			
Send SNMP trap			
Run script			
Create ServiceNow incident			
Send Syslog message			

10. References

Official Veeam Documentation:

https://helpcenter.veeam.com/docs/vbr/userguide/malware_detection.html?ver=13

Script to encrypt data:

<https://github.com/magnunscheffer/veeam/blob/main/demo-malware/encrypt.ps1>

Script to rename directories:

<https://github.com/magnunscheffer/veeam/blob/main/demo-malware/rename.sh>

Yara Rules Documentation:

<https://yara.readthedocs.io/en/stable/>

Yara Rule Eicar Example:

<https://github.com/magnunscheffer/veeam/blob/main/demo-malware/eicar.yara>

Examples of files used in fileserver:

<https://file-examples.com/index.php/text-files-and-archives-download/>